

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/260375733>

A single-session testing protocol to determine critical power and W'

Article in *Arbeitsphysiologie* · February 2014

DOI: 10.1007/s00421-014-2827-8 · Source: PubMed

CITATIONS

14

READS

2,747

3 authors:



Keren Constantini

Indiana University Bloomington

14 PUBLICATIONS 59 CITATIONS

[SEE PROFILE](#)



Surendran Sabapathy

Griffith University

113 PUBLICATIONS 1,332 CITATIONS

[SEE PROFILE](#)



Troy James Cross

Mayo Clinic - Rochester

47 PUBLICATIONS 255 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



Effect of laser PBM on muscle fatigue [View project](#)



The effects of shear stress on RBC biochemistry and haemorheology [View project](#)

A single-session testing protocol to determine critical power and W'

Keren Constantini · Surendran Sabapathy ·
Troy J. Cross

Received: 17 October 2013 / Accepted: 18 January 2014 / Published online: 22 February 2014
© Springer-Verlag Berlin Heidelberg 2014

Abstract

Purpose Critical power (CP), and the finite capacity to perform work above CP (W'), can be determined using a 3-min “all-out” cycling test (3MT). This protocol requires two laboratory visits: an incremental exercise test, followed by a 3MT on a separate day. The purpose of this study was to establish whether an incremental exercise test and a 3MT performed during a *single* laboratory visit can be used to accurately determine CP and W' .

Methods Twelve participants completed two experimental protocols: (1) *Combined protocol*: an incremental exercise test followed by a 3MT, with 20 min of recovery between exercise bouts; and (2) *Independent protocol*: the conventional 3MT protocol, performed on a separate day.

Results CP determined from the Combined (254 ± 117 W) and Independent (256 ± 118 W) protocols were not different ($p = 0.40$). Similarly, W' was not different ($p = 0.96$) between the Combined (13.7 ± 3.9 kJ) and Independent (13.7 ± 4.5 kJ) protocols. Linear regression revealed a strong level of measurement agreement between the protocols for CP and W' , evidenced by high R^2 values (≥ 0.85) and marginal standard errors of the estimates (CP = 5 W; $W' = 1.81$ kJ).

Conclusion A Combined protocol, consisting of an incremental exercise test followed by a 3MT, provides an accurate and valid method to determine an individual's CP and, to a lesser extent, W' . Furthermore, this protocol permits

the measurement of the gas-exchange threshold and peak O_2 uptake and, consequently, the moderate, heavy, and severe exercise-intensity domains may be defined within a *single* exercise-testing session.

Keywords Supramaximal exercise · Aerobic fitness · Cycling endurance

Abbreviations

3MT	3-Min “all-out” test
ATP	Adenosine triphosphate
CP	Critical power
P	Instantaneous cycling power
GET	Gas-exchange threshold
T_{lim}	Time to task-failure
O_2	Oxygen
$\dot{V}O_2$	O_2 uptake
$\dot{V}O_{2peak}$	Peak O_2 uptake
W'	Finite capacity to perform work above critical power

Introduction

Critical power (CP) represents the highest rate of muscular work where a metabolic ‘steady-state’ can be achieved (Poole et al. 1988; Jones et al. 2010; Vautier et al. 1995). According to the CP model proposed by Monod and Scherrer (1965), an individual may sustain exercise at intensities below CP (i.e. moderate and heavy-intensity domains) for an indefinite period of time, provided that sufficient stores of substrates for oxidative ATP resynthesis are available (Moritani et al. 1981; Monod and Scherrer 1965; Rossiter 2010). However, at work rates performed above CP (i.e. severe-intensity domain), exercise duration is determined by

Communicated by David C. Poole.

K. Constantini · S. Sabapathy · T. J. Cross (✉)
Heart Foundation Research Centre, Griffith Health Institute,
and the School of Allied Health Sciences, Gold Coast Campus,
Griffith University, Queensland 4222, Australia
e-mail: t.cross@griffith.edu.au

the rate of depletion of a finite energy reserve, termed the W' (Monod and Scherrer 1965; Moritani et al. 1981). This energy reserve is believed to represent the available anaerobic capacity and intramuscular O_2 stores, and its rate of depletion is dependent upon O_2 uptake kinetics and the rate of accumulation of metabolic by-products (Moritani et al. 1981; Jones et al. 2008; Miura et al. 1999, 2000; Burnley et al. 2011; Burnley and Jones 2007; Monod and Scherrer 1965; Murgatroyd et al. 2011). A number of studies have demonstrated that severe-intensity exercise can be maintained up until the point where W' is completely expended (Fukuba et al. 2003; Chidnok et al. 2012; Coats et al. 2003). Thereafter, muscular work can be sustained only at intensities below CP (Coats et al. 2003; Chidnok et al. 2013); that is, work rates that rely predominantly on energy provision from oxidative pathways (Moritani et al. 1981). In light of the above, it follows that CP is an important determinant of endurance performance (Jones et al. 2010; Smith et al. 1999; Vautier et al. 1995). Furthermore, this parameter, along with the W' , can be used to optimise training programmes and pacing strategies during cycling and running events (Clarke and Skiba 2013; Fukuba and Whipp 1999; Jones et al. 2010; Skiba et al. 2012; Vanhatalo et al. 2011).

CP is traditionally estimated by defining the relationship between power output (P) and exercise task-duration (T_{lim})—a relationship that is hyperbolic in nature (Monod and Scherrer 1965; Moritani et al. 1981; Poole et al. 1988). The asymptote of this hyperbola is CP, representing the power output at which task-duration tends to infinity. The finite area bracketed by the P - T_{lim} curve and its asymptote represents W' (Monod and Scherrer 1965; Hill 1993; Moritani et al. 1981). The traditional approach outlined above requires an individual to complete a minimum of 3–4 exhaustive, constant-load bouts of exercise on separate days (Monod and Scherrer 1965). Consequently, defining CP and W' in this manner is time consuming. Vanhatalo et al. (2007) recently demonstrated that CP can be accurately determined by averaging the power output attained over the final 30 s of a 3-min “all-out” cycling test (3MT). Additionally, W' may be determined from this test by calculating the finite work expended above CP. Subsequent investigations have established that the 3MT protocol provides a robust and valid estimate of the CP and W' (Johnson et al. 2011; Francis et al. 2010; Vanhatalo et al. 2008a, b).

The 3MT approach to determining CP and W' requires the application of a pre-determined linear resistance to the ergometer flywheel during the test (the “linear factor”). This “linear factor” is based on the power outputs attained at the gas-exchange threshold (GET) and peak O_2 uptake ($\dot{V}O_{2peak}$) (Vanhatalo et al. 2007), and therefore necessitates a prior incremental exercise test, typically performed on a separate day. In an effort to further reduce the number of testing sessions required to determine CP and W' , other investigators

have used a “linear factor” based on participants’ body mass, obviating a prior incremental exercise test (Bergstrom et al. 2012; Clark et al. 2012). While such findings are promising, these modified versions of the 3MT do not afford estimation of the GET and $\dot{V}O_{2peak}$ (Cross 2012). Thus, it may be argued that to define the boundaries of the heavy- and severe-intensity domains, and provide a comprehensive assessment of endurance ability, a 3MT protocol should be performed in conjunction with an incremental exercise test.

The aim of the present study was to determine whether CP and W' can be accurately determined using the 3MT within a *single* testing session; i.e. an incremental exercise test followed by a 3-min “all-out” cycling test, with a short period of recovery (20 min) separating the two tests. We hypothesised that CP and W' measured after incremental exercise, within a single testing session (Combined protocol), would not be different from that obtained on a separate day (Independent protocol).

Methods

Subjects and ethical approval

Twelve subjects (six women, six men; age: 26 ± 4 years, body mass: 71 ± 15 kg, height: 1.77 ± 0.11 m) volunteered to participate in this study and provided written, informed consent. The subjects underwent pre-participatory health screening to ensure they were non-smokers, with no history of cardiac, pulmonary or metabolic diseases. Subjects were recreationally active and accustomed to high-intensity exercise. Prior to each visit, subjects were instructed to avoid heavy exercise and alcohol consumption for 24 h, consumption of caffeinated drinks for 12 h, and intake of food for 3 h. Subjects were asked to maintain a similar diet during the 24 h before each test. The present study conformed to the principles outlined in the Declaration of Helsinki and was approved by the Griffith University Human Research Ethics Committee.

Experimental design

Subjects attended the laboratory on two occasions, separated by at least 48 h of rest. During the first testing session, subjects completed a ramp incremental cycling test (Ramp) to volitional exhaustion followed by 20 min of recovery (5 min active recovery + 15 min passive recovery). After the 20-min recovery period, subjects completed a 3MT (Combined Protocol). On a separate day, participants performed a 3MT without prior incremental exercise (Independent Protocol). Studies have shown that the 3MT, used here in the Independent protocol, yields a valid and reliable estimate of CP and W' , respectively (Johnson et al. 2011; Francis et al. 2010; Vanhatalo et al. 2007, 2008a, b).

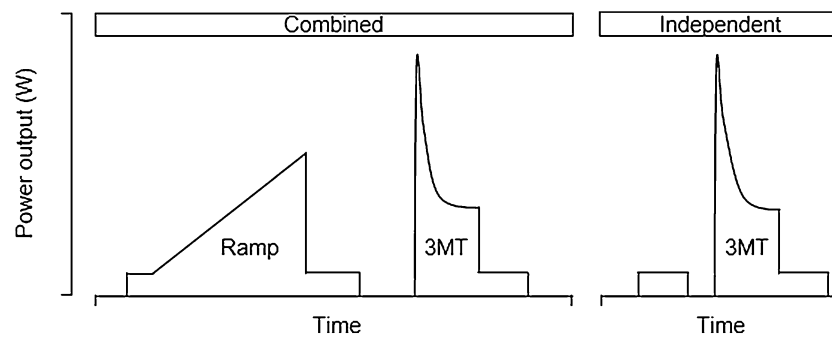


Fig. 1 Experimental design for the Combined and Independent protocols. The Combined protocol consisted of a ramp incremental cycling test (Ramp) to volitional exhaustion and a 3-min “all-out” test

(3MT). The tests were separated by a 20-min recovery (5 active + 15 passive). During the Independent protocol, performed on a separate day, participants completed a 3MT without prior ramp exercise

Therefore, for the purposes of this investigation, the Independent protocol provides the “criterion” measures of CP and W' . The Combined and Independent protocols are illustrated in Fig. 1.

Incremental exercise test

Subjects performed a ramped incremental exercise (Ramp) test on an electrically braked cycle ergometer (Excalibur Sport, Lode BV, Groningen, Netherlands). The Ramp test commenced with a 3-min constant-load warm-up period of cycling at 0–90 W. Thereafter, power output was continuously increased at an overall rate of 25 and 30 W min^{-1} for female and male subjects, respectively. The power output used during the warm-up was determined a priori based on each subject's training status, such that time-to-exhaustion during the incremental exercise test would be 8–12 min. A self-selected pedal cadence ($84 \pm 8 \text{ rev min}^{-1}$) was maintained by each subject until volitional exhaustion. The Ramp test was terminated when subjects could not continue cycling at their self-selected cadence despite strong verbal encouragement; i.e. when cadence fell $>10 \text{ rev min}^{-1}$ below the self-selected cadence. After the Ramp test was terminated, the workload was immediately decreased to 30 W and subjects cycled for a further 5 min (active recovery period). The active recovery was followed by 15 min of passive recovery. Cardiac rhythm and heart rate were monitored continuously during all testing procedures using a 5-lead electrocardiogram (X-Scribe, Mortara Instrument, Inc., Milwaukee, WI). Gas-exchange variables were measured breath-by-breath (Ultima Cardio O_2 , Medical Graphics Corporation, St. Paul, MN, USA). The O_2 and CO_2 analysers were calibrated before each test using room air and a calibration gas (12 % O_2 ; 5 % CO_2 ; balanced N_2). Subjects breathed through a mouthpiece attached to a bi-directional differential pressure pneumotachograph (pre-Vent, Medical Graphics Corporation, St. Paul, MN, USA) while wearing a nose-clip. The pneumotachograph was calibrated at varying

flows (30–360 L min^{-1}) using a 3-L syringe. Peak O_2 uptake ($\dot{V}\text{O}_{2\text{peak}}$) was determined as the highest O_2 uptake ($\dot{V}\text{O}_2$) value for 30-s bin-averaged data measured during the Ramp test, and peak power output was determined as the final work rate attained during ramp exercise. The GET was estimated using the modified V-slope method after the data were reduced to 15-s intervals (Schneider et al. 1993).

3-Min “all-out” cycling test (3MT)

The 3MT was performed on an electronically braked cycle ergometer (Excalibur Sport, Lode BV, Groningen, Netherlands), using the procedures described by Burnley et al. (2006b) and Vanhatalo et al. (2007). The test commenced with 3 min of unloaded cycling using the cadence selected during the Ramp test (Vanhatalo et al. 2007). During the last 10 s of the 3-min unloaded cycling period, subjects were instructed to increase their pedal cadence to ~ 110 – 120 rev min^{-1} . Upon commencing the 3MT, subjects rapidly increased their pedal cadence with the intention of “cycling as fast as possible”. To ensure maximal effort, subjects were instructed to maintain their cadence as high as possible throughout the 3-min duration of the test (Burnley et al. 2006b; Vanhatalo et al. 2007). Strong verbal encouragement was provided by the two investigators during the entire 3MT. To discourage pacing during the test, subjects were not given feedback on their pedal cadence or exercise time. The operating mode of the cycle ergometer during the 3MT was set to “linear”, where the resistance applied to the ergometer flywheel was proportional to the product of pedal cadence and an inputted “linear factor”. The “linear factor” was determined as $\text{power cadence}^{-2}$, such that the subjects would attain a power output equal to a work rate halfway between the GET and $\dot{V}\text{O}_{2\text{peak}}$ ($\Delta 50$ %) when cycling at their self-selected pedal cadence (Vanhatalo et al. 2007; Burnley et al. 2006b).

CP was calculated as the average power output attained during the final 30 s of the 3MT. The parameter W' was determined as the finite amount of external work (in kJ)

expended above CP during the 3MT. Gas-exchange data were obtained during the 3MT using the same methods as described for the incremental exercise test. $\dot{V}O_{2\text{peak}}$ during the 3MT was determined as the highest 30-s average $\dot{V}O_2$ value observed during the test.

The 3MT completed during the Combined protocol was performed immediately after the 20-min recovery period following the Ramp test. Subjects did not consume food or carbohydrate beverages during this recovery period. During the Independent protocol, subjects first completed a 5-min warm-up at a low exercise-intensity that was below the GET (60–100 W). The warm-up was followed by 5-min of passive rest, after which subjects mounted the cycle ergometer and performed the 3MT. As with the Combined protocol, this 3MT was preceded with 3-min unloaded cycling using the self-selected cadence during the Ramp test (Burnley et al. 2006b; Vanhatalo et al. 2007). The “linear factor” used during the Independent protocol was identical to that used during the 3MT performed during the Combined protocol. The CP and W' values obtained from the Combined and Independent protocols are notated here as CP_{COMB} and W'_{COMB} , and $CP_{3\text{MT}}$ and $W'_{3\text{MT}}$, respectively.

Statistical analyses

Paired samples *t*-tests were used to compare CP and W' values obtained during the Combined and Independent protocols. Repeated measures one-way analysis of variance was used to compare peak $\dot{V}O_2$ values attained during the Ramp test and the 3MTs completed as part of the Combined and Independent protocols, with pair-wise comparisons assessed using the Bonferroni post hoc adjustment. The absolute agreement between CP and W' determined from Combined and Independent protocols were assessed using the standard errors of the estimates (SEE) obtained via linear regression (Hopkins 2000)—the SEE is also known as the *typical error*. The relative variability in CP and W' between protocols was evaluated using the coefficient of variation (CV; Hopkins 2000). Results are presented as mean $\pm$ SD unless otherwise stated. All data were analysed using SPSS 20.0 (SPSS, Inc., Chicago, IL, USA). Statistical analyses were considered significant if $p < 0.05$.

Results

Incremental exercise test

The $\dot{V}O_{2\text{peak}}$ and peak power output achieved during the Ramp incremental test were $3.70 \pm 1.40 \text{ L min}^{-1}$ ($51.0 \pm 11.9 \text{ mL kg}^{-1} \text{ min}^{-1}$) and $346 \pm 117 \text{ W}$, respectively. Peak exercise heart rate was $186 \pm 13 \text{ beats min}^{-1}$, representing $96 \pm 7 \%$ of age-predicted maximum.

The GET occurred at a $\dot{V}O_2$ of $2.22 \pm 0.96 \text{ L min}^{-1}$ ($30.6 \pm 10.0 \text{ mL kg}^{-1} \text{ min}^{-1}$; $59 \pm 7 \%$ $\dot{V}O_{2\text{peak}}$) and a power output of $171 \pm 80 \text{ W}$ ($47 \pm 9 \%$ peak power output). The $\Delta 50 \%$ power output used to compute each subject's “linear factor” for the 3MT protocol was $259 \pm 98 \text{ W}$ ($74 \pm 4 \%$ peak power output).

Ramp versus 3MT

The $\dot{V}O_{2\text{peak}}$ attained during the 3MTs performed in the Combined and Independent protocols were $52.5 \pm 11.1 \text{ mL kg min}^{-1}$ and $49.2 \pm 11.9 \text{ mL kg min}^{-1}$, respectively. $\dot{V}O_{2\text{peak}}$ measured during the Combined and Independent protocols represented 103 ± 6 and $96 \pm 5 \%$ of the Ramp $\dot{V}O_{2\text{peak}}$, respectively. There was a significant main effect of test protocol on $\dot{V}O_{2\text{peak}}$ ($F = 6.84$; $p < 0.05$). Post hoc analyses revealed that there was no difference between $\dot{V}O_{2\text{peak}}$ attained during the Ramp test and either of the two 3MT protocols. However, $\dot{V}O_{2\text{peak}}$ attained during the Independent protocol was lower ($p < 0.05$) than that achieved during the Combined protocol. Peak exercise heart rate during the Ramp test was significantly higher ($p < 0.05$) when compared to both the Combined ($179 \pm 13 \text{ beats min}^{-1}$) and Independent ($176 \pm 15 \text{ beats min}^{-1}$) protocols.

Combined versus Independent protocols

Figure 2 displays the group-averaged and mean differences in power output during the 3MT in the Combined and Independent protocols. The individual and mean differences in CP and W' between the Combined and Independent protocols are presented in Fig. 3. CP determined during the Combined (CP_{COMB} ; $254 \pm 117 \text{ W}$) and Independent ($CP_{3\text{MT}}$; $256 \pm 118 \text{ W}$) protocols were not different ($p = 0.40$). The coefficient of determination for the linear relationship between CP_{COMB} and $CP_{3\text{MT}}$ was very strong ($R^2 = 0.998$; Fig. 4, Panel a). W'_{COMB} ($13.7 \pm 3.9 \text{ kJ}$) and $W'_{3\text{MT}}$ ($13.7 \pm 4.5 \text{ kJ}$) were not significantly different ($p = 0.96$). Moreover, the coefficient of determination for the relationship between W'_{COMB} and $W'_{3\text{MT}}$ was strong ($R^2 = 0.853$; Fig. 4, Panel b). The relative variability (i.e. CV) for CP and W' between the Combined and Independent protocols was 3.5 and 12.0 %, respectively. The peak power output attained during the 3MT in the Combined protocol ($821 \pm 327 \text{ W}$) was significantly lower ($p < 0.05$) than that achieved in the Independent 3MT ($889 \pm 368 \text{ W}$).

Discussion

We sought to determine whether an incremental exercise test and a 3MT protocol may be completed within a single

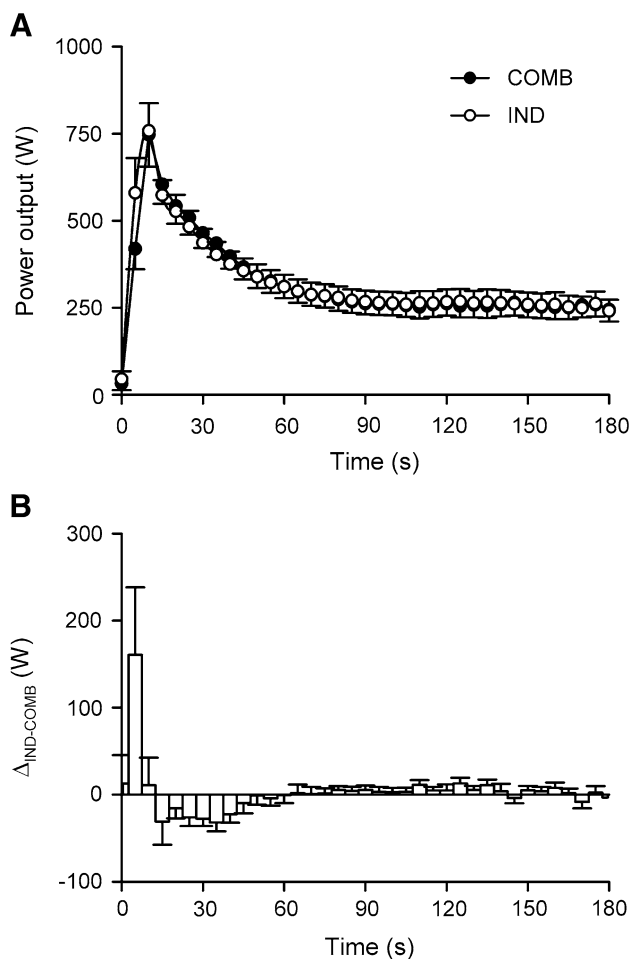


Fig. 2 Group averaged and mean differences in power output during the Combined and Independent 3-min “all-out” tests (3MT). The closed circles denote values obtained from the Combined 3MT (COMB). The open circles represent the Independent 3MT (IND) measurements. Note the similarity in power outputs between the two 3MTs between 150 and 180 s (representing critical power). In panel b, a negligible mean difference (represented by the small bar graphs) between the power outputs during the final 30 s of the Combined and Independent 3MT is observed

testing session, without compromising the accuracy with which CP and W' are determined. The novel finding of the present study was that CP and W' were not significantly affected by prior exhaustive, incremental exercise. That is, CP and W' determined during the Combined protocol displayed a strong level of measurement agreement with those obtained from the criterion test, the Independent protocol. Furthermore, while $\dot{V}O_{2peak}$ attained during the 3MT in the Combined protocol was significantly greater than that achieved in the Independent protocol, these values were not significantly different from peak $\dot{V}O_2$ measured during the Ramp test.

The differences in the $\dot{V}O_{2peak}$ responses during the 3MTs may be related to a priming effect of the prior

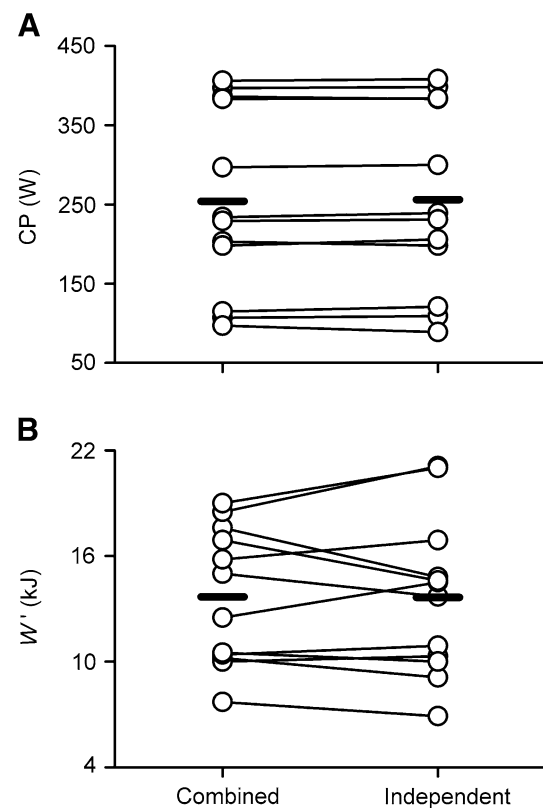


Fig. 3 Individual and mean differences between the Combined and Independent 3-min “all-out” tests for CP and W' . CP critical power; W' finite capacity to perform work above CP. Solid lines represent the group averages. Both CP and W' determined from the two protocols were not different

incremental exercise (i.e. in the Combined protocol). Although Vanhatalo and Jones (2009) reported no differences in peak $\dot{V}O_2$ values during a 3MT performed with prior sprint exercise, Sahlin et al. (2005) showed that priming exercise reduced gross efficiency during a subsequent high-intensity exercise bout. Furthermore, Burnley et al. (2011) demonstrated that the peak $\dot{V}O_2$ attained during a range of constant work rate exercise bouts (~70–100 % of peak work rate), performed to exhaustion and preceded by severe-intensity priming exercise, was increased when compared to the same exercise bouts performed without prior exercise. Thus, the O_2 cost of cycling during the 3MT performed in the Combined protocol in our study may have been higher than in the Independent protocol because of the preceding Ramp test. Nevertheless, the peak $\dot{V}O_2$ attained during a 3MT is known to vary with respect to $\dot{V}O_{2peak}$ determined from incremental exercise: Burnley et al. (2006b) demonstrated that the peak $\dot{V}O_2$ attained during a 3MT was similar to a ramp $\dot{V}O_{2peak}$, while Vanhatalo et al. (2007) reported peak $\dot{V}O_2$ during a 3MT to be slightly, but significantly, lower than ramp $\dot{V}O_{2peak}$ (~97 %).

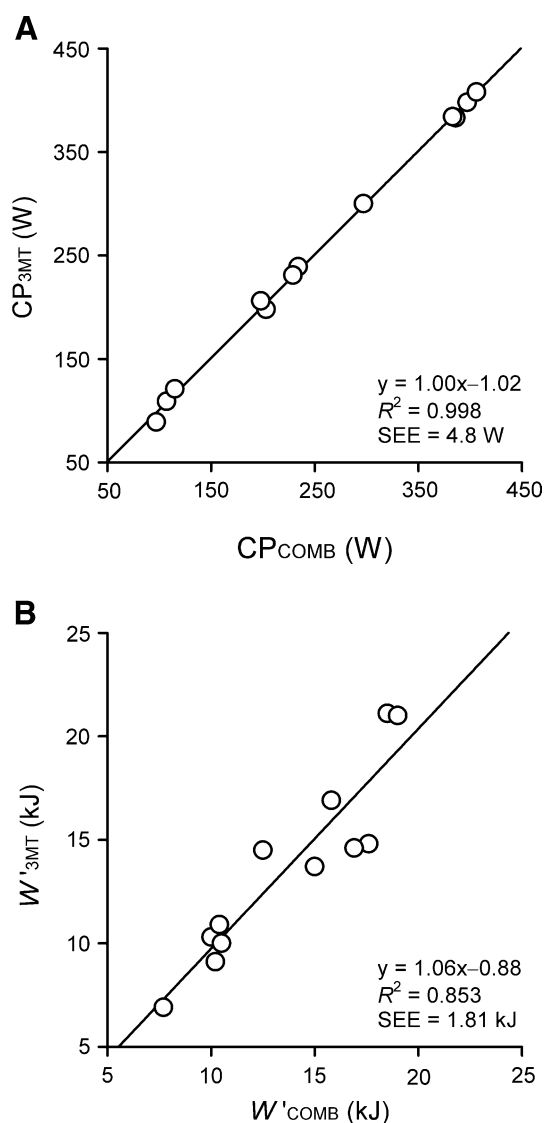


Fig. 4 The absolute agreement between the Combined and Independent 3-min “all-out” tests for critical power (CP_{COMB} vs. CP_{3MT}) and W' (W'_{COMB} vs. W'_{3MT}) measurements. CP critical power, W' finite capacity to perform work above CP, R^2 coefficient of determination, SEE standard error of the estimates

In the present study, we have shown that CP determined from the 3MT during the Combined protocol was practically equivalent to that obtained from the Independent protocol (Fig. 4). The observation that the preceding Ramp exercise did not impact detrimentally on CP during the subsequent 3MT is not surprising as it is well documented that CP is unaffected by a prior bout of exercise, even when the bout is exhaustive (Ferguson et al. 2007, 2010; Burnley et al. 2006a, 2011; Simpson et al. 2012). Moreover, the typical error associated with CP_{COMB} was <5 W. This high level of measurement accuracy was observed across a wide range of measured CP (range 89–408 W) and $\dot{V}O_{2\text{peak}}$ values (range 39.9–70.1 mL kg⁻¹ min⁻¹). Accordingly,

we reason that the Combined protocol provides as robust and accurate an estimation of CP as the Independent 3MT, regardless of training status and aerobic power.

W' was also similar between the Combined and Independent protocols, despite a significantly higher peak power output attained in the latter. Nevertheless, Fig. 4 and Table 1 reveal that W'_{COMB} produced a proportional, although biased estimate of the participants' “true” W' . That is, W'_{COMB} marginally underestimated W'_{3MT} when the observed W'_{COMB} was smaller than 14.1 kJ; the converse being true when W'_{COMB} was greater than 14.1 kJ. Notwithstanding this proportional bias, the typical error associated with W'_{COMB} was 1.81 kJ. At present, there is little consensus on what an acceptable measurement error is when estimating the W' via the 3MT. Determination of W' from the Combined protocol was shown to be associated with a similar amount of measurement error (range 1.46–1.95 kJ) to that reported by previous investigators using either the conventional or modified versions of the 3MT (Bergstrom et al. 2012; Burnley et al. 2006b; Clark et al. 2012; Johnson et al. 2011).

A further consideration regarding the accuracy of W'_{COMB} relates to the duration of the recovery period between the Ramp and 3MT performed in the Combined protocol. In contrast to CP, the effect of prior exercise on W' appears related to the duration and intensity of the preceding exercise bout, and the duration of recovery between exercise bouts (Burnley et al. 2011; Ferguson et al. 2010; Vanhatalo and Jones 2009). Ferguson et al. (2010) showed that 15 min of active recovery following exhaustive constant work rate exercise resulted in approximately 86 % recovery of control W' values, while others have shown that W' appears fully recovered approximately 10 min following non-exhaustive supra-CP exercise (Burnley et al. 2011; Vanhatalo and Jones 2009). Practically, we chose 20 min of recovery between the Ramp and 3MT as an expedient duration because pilot work indicated this time frame was sufficient to process the Ramp tests results (determine the power outputs attained at the GET and peak exercise), compute the linear factor, recalibrate equipment and programme the ergometer protocol. Consequently, we considered 20-min a reasonable and practical duration of recovery to test our hypotheses. However, Skiba et al. (2012) recently described the time course of W' reconstitution as an exponential function, yielding a time constant of ~377 s and implying a full recovery duration of approximately 25 min. This finding suggests that the recovery period we implemented may have resulted in the underestimation of W'_{COMB} in some participants, and a longer recovery duration between the Ramp and 3MT may deserve consideration in future applications of the Combined protocol. Nevertheless, considering that our range of measurement errors is comparable to that reported by others (see above), the Combined protocol appears to produce an acceptable estimate of W' .

Table 1 Comparison of the present findings with those reported by Bergstrom et al. (2012) and Clark et al. (2012)

Source of data	<i>n</i>	Linear factor	CP			<i>W'</i>		
			Equation	<i>R</i> ²	SEE (W)	Equation	<i>R</i> ²	SEE (kJ)
Bergstrom et al. (2012)	12	4.5 %BM	1.16 CP _{%BM} − 22.2	0.90	18.3	0.52 <i>W'</i> _{%BM} + 5.80	0.61	1.93
Clark et al. (2012)	15	3, 4 and 5 %BM	1.01 CP _{%BM} − 1.54	0.94	10.4	0.32 <i>W'</i> _{%BM} + 5.12	0.21	1.95
Constantini et al.	12	Δ50 % cadence ^{−2}	1.00 CP _{COMB} − 1.02	0.99	4.8	1.06 <i>W'</i> _{COMB} − 0.88	0.85	1.81

The regression analyses presented in this Table were performed on the individual data reported by Bergstrom et al. (2012) and Clark et al. (2012). Note that the SEE obtained from linear regression analysis may also be referred to as the typical error of measurement

%BM percentage body mass, Δ50 % median work rate between the gas-exchange threshold and peak O₂ uptake, CP critical power, *W'* finite capacity to perform above CP, CP_{%BM} and *W'*_{%BM} CP and *W'* obtained using %BM as a “linear factor”, CP_{COMB} and *W'*_{COMB} CP and *W'* obtained from the Combined protocol developed in the present study, cadence self-selected pedalling frequency during incremental exercise test, *R*² coefficient of determination, SEE standard error of the estimate

It is generally recommended that a preliminary 3MT be completed on a separate occasion prior to the 3MT of interest (Jones et al. 2010; Vanhatalo et al. 2007). This familiarisation trial is deemed necessary to ensure that valid estimates of CP and *W'* are obtained during the experimental 3MT (Vanhatalo et al. 2007; Jones et al. 2010; Burnley et al. 2006b). The current study design did not include a familiarisation trial, because it would have required an additional visit to the laboratory. The extra visit would have detracted from the central aim of this study: to develop an expeditious cycling protocol where CP and *W'* are determined during a single laboratory visit. The conscious decision not to include a familiarisation trial, particularly before the Combined protocol, also precluded the randomisation of the two test protocols. The Combined test was always performed first, because the parameters used to compute the “linear factor” for the 3MT in both protocols were obtained from the Ramp test (performed in the Combined protocol). Nevertheless, a strong level of measurement agreement in CP and *W'* values was observed between experimental protocols, and the CV values for CP from the current study (3.5 %) were similar to those obtained from repeated 3MTs performed *after* subjects have completed a familiarisation trial (CV = 3.0 %; Burnley et al. 2006b). Furthermore, in a recent test–retest reliability investigation of the 3MT, Johnson et al. (2011) reported a CV of 21 % for *W'*, almost twice the magnitude found in the current study (12 %). For the reasons given above, it would appear that a familiarisation trial is not necessary to yield accurate estimates of CP and *W'* using the Combined protocol, and that a lack of randomisation did not confound the data obtained from the current study. The fact that the peak $\dot{V}O_2$ attained during both the Combined and Independent protocols was greater than 95 % of the Ramp $\dot{V}O_{2peak}$ (and that no decline in $\dot{V}O_2$ was observed throughout the 3MTs), provides further support for our conviction that all 3MT protocols completed in the present study were valid (Jones et al. 2010).

It must be acknowledged that we are not the first to estimate the CP and *W'* within a *single* laboratory visit. Recently, Bergstrom et al. (2012) and Clark et al. (2012) have shown that CP and *W'* can be determined using a modified version of the 3MT, where the “linear factor” is set equal to a percentage of participants’ body mass instead of the conventional value of Δ50 % cadence^{−2} (see “Methods”). This modified 3MT obviates the need for a preliminary incremental exercise test, and therefore reduces the total number of visits to a single testing session. In order to compare our findings with those of previous investigators, linear regression analyses were performed on the individual data reported by Bergstrom et al. (2012) and Clark et al. (2012) and presented in Table 1. It may be argued that for a modified-3MT protocol to be considered a valid surrogate for the conventional “all-out” test, the regression equations presented in Table 1 should (1) produce a linear function close to that of the line of identity ($y = 1.00x$), i.e. generate the least amount of proportional bias; (2) the estimator should predict the criterion with a minimal typical error as given by the SEE; and (3) display a high degree of “explanatory power” as indicated by the *R*². Based on these criteria, it would appear that our Combined protocol yields the most accurate determination of an individual’s “true” CP and *W'* out of all currently available “single-session” protocols. It is emphasised, however, that the above analyses were performed using data obtained from separate investigations and, subsequently, the conclusions drawn from these results must be regarded as strictly informal.

Practical implications

By definition, CP represents the upper limit of sustainable muscular work (Monod and Scherrer 1965; Poole et al. 1988; Hill 1993) and, as such, may be regarded as an important determinant of endurance performance ability (Burnley et al. 2006b; Jones et al. 2010; Vautier et al. 1995; Smith et al. 1999). In addition to the above, there is growing interest in using an athlete’s CP and *W'* to optimise

training programmes and devise pacing strategies during competitive cycling events (Jones et al. 2010; Vanhatalo et al. 2011; Fukuba and Whipp 1999; Clarke and Skiba 2013; Skiba et al. 2012). These points do not, however, displace the importance of determining other key parameters of aerobic fitness (Saltin and Astrand 1967; Jones and Carter 2000; Coyle et al. 1991). Given that the upper boundaries of the moderate, heavy and severe domains are demarcated by the GET, CP and $\dot{V}O_{2peak}$, respectively (Poole et al. 1988; Rossiter 2010; Stienacker and Ward 1996), it follows that determination of these parameters is necessary to comprehensively profile the endurance ability of an athlete. Consequently, a distinct advantage of our Combined protocol over other “single-session” designs (Clark et al. 2012; Bergstrom et al. 2012) is that it allows for the determination of not only CP and W' , but also the GET and $\dot{V}O_{2peak}$. To the best of our knowledge, this study is the first to demonstrate that the boundaries of the moderate-, heavy- and severe-intensity domains may be accurately determined within a *single* visit to the laboratory.

Conclusions

A Combined protocol, consisting of a Ramp incremental exercise test followed by a 3MT, provides an accurate and valid method to determine an individual's CP and W' . Furthermore, performing this sequence of tests allows the definition of the boundaries of the moderate-, heavy- and severe-intensity domains within a single visit to the laboratory. Future studies should investigate the repeatability and robustness of CP and W' values obtained using the Combined protocol.

Acknowledgments The Authors thank each participant for their commitment and efforts in this study. TJC was supported by a Griffith University Postdoctoral Fellowship.

Conflict of interest The authors declare that they have no conflict of interest.

Ethical standards The experiments performed in this study complied with the current laws of Australia. The present study conformed to the principles outlined in the Declaration of Helsinki and was approved by the Griffith University Human Research Ethics Committee.

References

- Bergstrom HC, Housh TJ, Zuniga JM, Camic CL, Traylor DA, Schmidt RJ, Johnson GO (2012) A new single work bout test to estimate critical power and anaerobic work capacity. *J Strength Cond Res* 26(3):656–663. doi:10.1519/JSC.0b013e31822b7304
- Burnley M, Jones AM (2007) Oxygen uptake kinetics as a determinant of sports performance. *Eur J Sport Sci* 7(2):63–79. doi:10.1080/17461390701456148
- Burnley M, Doust JH, Jones AM (2006a) Time required for the restoration of normal heavy exercise VO_2 kinetics following prior heavy exercise. *J Appl Physiol* 101(5):1320–1327. doi:10.1152/jappphysiol.00475.2006
- Burnley M, Doust JH, Vanhatalo A (2006b) A 3-min all-out test to determine peak oxygen uptake and the maximal steady state. *Med Sci Sports Exerc* 38(11):1995–2003. doi:10.1249/01.mss.0000232024.06114.a6
- Burnley M, Davison G, Baker JR (2011) Effects of priming exercise on VO_2 kinetics and the power-duration relationship. *Med Sci Sports Exerc* 43(11):2171–2179. doi:10.1249/MSS.0b013e31821ff26d
- Chidnok W, Dimenna FJ, Bailey SJ, Vanhatalo A, Morton RH, Wilkerson DP, Jones AM (2012) Exercise tolerance in intermittent cycling: application of the critical power concept. *Med Sci Sports Exerc* 44(5):966–976. doi:10.1249/MSS.0b013e31823ea28a
- Chidnok W, Fulford J, Bailey SJ, Dimenna FJ, Skiba PF, Vanhatalo A, Jones AM (2013) Muscle metabolic determinants of exercise tolerance following exhaustion: relationship to the ‘critical power’. *J Appl Physiol*. doi:10.1152/jappphysiol.00334.2013
- Clark IE, Murray SR, Pettitt RW (2012) Alternative procedures for the 3-min all-out exercise test. *J Strength Cond Res* 27(8):2104–2112. doi:10.1519/JSC.0b013e3182785041
- Clarke DC, Skiba PF (2013) Rationale and resources for teaching the mathematical modeling of athletic training and performance. *Adv Physiol Educ* 37(2):134–152. doi:10.1152/advan.00078.2011
- Coats EM, Rossiter HB, Day JR, Miura A, Fukuba Y, Whipp BJ (2003) Intensity-dependent tolerance to exercise after attaining VO_{2max} in humans. *J Appl Physiol* 95(2):483–490. doi:10.1152/jappphysiol.01142.2002
- Coyle EF, Feltner ME, Kautz SA, Hamilton MT, Montain SJ, Baylor AM, Abraham LD, Petrek GW (1991) Physiological and biomechanical factors associated with elite endurance cycling performance. *Med Sci Sports Exerc* 23(1):93–107
- Cross TJ (2012) Determination of critical power and anaerobic work capacity: is one test better than two? *J Strength Cond Res*. doi:10.1519/JSC.0b013e31826ee4e1
- Ferguson C, Whipp BJ, Cathcart AJ, Rossiter HB, Turner AP, Ward SA (2007) Effects of prior very-heavy intensity exercise on indices of aerobic function and high-intensity exercise tolerance. *J Appl Physiol* 103(3):812–822. doi:10.1152/jappphysiol.01410.2006
- Ferguson C, Rossiter HB, Whipp BJ, Cathcart AJ, Murgatroyd SR, Ward SA (2010) Effect of recovery duration from prior exhaustive exercise on the parameters of the power-duration relationship. *J Appl Physiol* 108(4):866–874. doi:10.1152/jappphysiol.91425.2008
- Francis JT Jr, Quinn TJ, Amann M, LaRoche DP (2010) Defining intensity domains from the end power of a 3-min all-out cycling test. *Med Sci Sports Exerc* 42(9):1769–1775. doi:10.1249/MSS.0b013e3181d612e8
- Fukuba Y, Whipp BJ (1999) A metabolic limit on the ability to make up for lost time in endurance events. *J Appl Physiol* 87(2):853–861
- Fukuba Y, Miura A, Endo M, Kan A, Yanagawa K, Whipp BJ (2003) The curvature constant parameter of the power-duration curve for varied-power exercise. *Med Sci Sports Exerc* 35(8):1413–1418. doi:10.1249/01.MSS.0000079047.84364.70
- Hill DW (1993) The critical power concept. A review. *Sports Med* 16(4):237–254
- Hopkins WG (2000) Measures of reliability in sports medicine and science. *Sports Med* 30(1):1–15
- Johnson TM, Sexton PJ, Placek AM, Murray SR, Pettitt RW (2011) Reliability analysis of the 3-min all-out exercise test for cycle ergometry. *Med Sci Sports Exerc* 43(12):2375–2380. doi:10.1249/MSS.0b013e318224cb0f

- Jones AM, Carter H (2000) The effect of endurance training on parameters of aerobic fitness. *Sports Med* 29(6):373–386
- Jones AM, Wilkerson DP, DiMenna F, Fulford J, Poole DC (2008) Muscle metabolic responses to exercise above and below the “critical power” assessed using ³¹P-MRS. *Am J Physiol Regul Integr Comp Physiol* 294(2):R585–R593. doi:[10.1152/ajpr.egu.00731.2007](https://doi.org/10.1152/ajpr.egu.00731.2007)
- Jones AM, Vanhatalo A, Burnley M, Morton RH, Poole DC (2010) Critical power: implications for determination of $\dot{V}O_{2\max}$ and exercise tolerance. *Med Sci Sports Exerc* 42(10):1876–1890. doi:[10.1249/MSS.0b013e3181d9cf7f](https://doi.org/10.1249/MSS.0b013e3181d9cf7f)
- Miura A, Kino F, Kajitani S, Sato H, Fukuba Y (1999) The effect of oral creatine supplementation on the curvature constant parameter of the power-duration curve for cycle ergometry in humans. *Jpn J Physiol* 49(2):169–174
- Miura A, Sato H, Sato H, Whipp BJ, Fukuba Y (2000) The effect of glycogen depletion on the curvature constant parameter of the power-duration curve for cycle ergometry. *Ergonomics* 43(1):133–141. doi:[10.1080/001401300184693](https://doi.org/10.1080/001401300184693)
- Monod H, Scherrer J (1965) The work capacity of a synergic muscular group. *Ergonomics* 8(1–4):329–338
- Moritani T, Nagata A, deVries HA, Muro M (1981) Critical power as a measure of physical work capacity and anaerobic threshold. *Ergonomics* 24(5):339–350
- Murgatroyd SR, Ferguson C, Ward SA, Whipp BJ, Rossiter HB (2011) Pulmonary O_2 uptake kinetics as a determinant of high-intensity exercise tolerance in humans. *J Appl Physiol* 110(6):1598–1606. doi:[10.1152/jappphysiol.01092.2010](https://doi.org/10.1152/jappphysiol.01092.2010)
- Poole DC, Ward SA, Gardner GW, Whipp BJ (1988) Metabolic and respiratory profile of the upper limit for prolonged exercise in man. *Ergonomics* 31(9):1265–1279. doi:[10.1080/00140138808966766](https://doi.org/10.1080/00140138808966766)
- Rossiter HB (2010) Exercise: kinetic considerations for gas exchange. *comprehensive physiology*. John Wiley & Sons, Inc. doi:[10.1002/cphy.c090010](https://doi.org/10.1002/cphy.c090010)
- Sahlin K, Sorensen JB, Gladden LB, Rossiter HB, Pedersen PK (2005) Prior heavy exercise eliminates $\dot{V}O_2$ slow component and reduces efficiency during submaximal exercise in humans. *J Physiol* 564(Pt 3):765–773. doi:[10.1113/jphysiol.2005.083840](https://doi.org/10.1113/jphysiol.2005.083840)
- Saltin B, Astrand PO (1967) Maximal oxygen uptake in athletes. *J Appl Physiol* 23(3):353–358
- Schneider DA, Phillips SE, Stoffolano S (1993) The simplified V-slope method of detecting the gas exchange threshold. *Med Sci Sports Exerc* 25(10):1180–1184
- Simpson PL, Jones AM, Vanhatalo A, Wilkerson DP (2012) Influence of initial metabolic rate on the power-duration relationship for all-out exercise. *Eur J Appl Physiol* 112(7):2467–2473. doi:[10.1007/s00421-011-2214-7](https://doi.org/10.1007/s00421-011-2214-7)
- Skiba PF, Chidnok W, Vanhatalo A, Jones AM (2012) Modeling the expenditure and reconstitution of work capacity above critical power. *Med Sci Sports Exerc* 44(8):1526–1532. doi:[10.1249/MS.S.0b013e3182517a80](https://doi.org/10.1249/MS.S.0b013e3182517a80)
- Smith JC, Dangelmaier BS, Hill DW (1999) Critical power is related to cycling time trial performance. *Int J Sports Med* 20(6):374–378. doi:[10.1055/s-2007-971147](https://doi.org/10.1055/s-2007-971147)
- Steinacker JrM, Ward SA (1996) The physiology and pathophysiology of exercise tolerance. Plenum Press, New York
- Vanhatalo A, Jones AM (2009) Influence of prior sprint exercise on the parameters of the ‘all-out critical power test’ in men. *Exp Physiol* 94(2):255–263. doi:[10.1113/expphysiol.2008.045229](https://doi.org/10.1113/expphysiol.2008.045229)
- Vanhatalo A, Doust JH, Burnley M (2007) Determination of critical power using a 3-min all-out cycling test. *Med Sci Sports Exerc* 39(3):548–555. doi:[10.1249/mss.0b013e31802dd3e6](https://doi.org/10.1249/mss.0b013e31802dd3e6)
- Vanhatalo A, Doust JH, Burnley M (2008a) A 3-min all-out cycling test is sensitive to a change in critical power. *Med Sci Sports Exerc* 40(9):1693–1699. doi:[10.1249/MSS.0b013e318177871a](https://doi.org/10.1249/MSS.0b013e318177871a)
- Vanhatalo A, Doust JH, Burnley M (2008b) Robustness of a 3 min all-out cycling test to manipulations of power profile and cadence in humans. *Exp Physiol* 93(3):383–390. doi:[10.1113/expphysiol.2007.039883](https://doi.org/10.1113/expphysiol.2007.039883)
- Vanhatalo A, Jones AM, Burnley M (2011) Application of critical power in sport. *Int J Sports Physiol Perform* 6(1):128–136
- Vautier JF, Vandewalle H, Arabi H, Monod H (1995) Critical power as an endurance index. *Appl Ergon* 26(2):117–121